Tim Csernica

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Education

California Institute of Technology Pasadena, CA
PhD in Chemistry June 2024

- Advisor: John Eiler
- Committee: Geoff Blake, Brian Stoltz, & Rudy Marcus
- Thesis: Exploring Nature's Fingerprints with Isotopic Distributions

University of Chicago Chicago, IL
B.S. with Honors in Chemistry June 2018

- Advisor: Steven Sibener
- Phi Beta Kappa

Professional Experience

2025-Present	**Postdoctoral Scholar,** Civil and Environmental Engineering, Northwestern University
2024-2025	**Staff Scientist (Remote)**, Geological and Planetary Sciences, Caltech
2024	**Postdoctoral Scholar**, Geological and Planetary Sciences, Caltech
2018-2024	**Graduate Researcher**, Chemistry and Chemical Engineering, Caltech
2017	**Undergraduate Researcher**, Chemical Engineering, Technion – Israel Institute of Tech.
2015-2018	**Undergraduate Researcher**, Chemistry, University of Chicago
2015	**University of Chicago Mathematics REU**, Mathematics, University of Chicago

Grants and Awards

2024	**Merkin Innovation Seed Grant:** *Natural Abundance Stable Isotopes of Metabolites for Cancer Diagnosis* (non-PI)	*$100,000*
2021	**Distinguished Alumnus: Lewisburg Area High School**	-
2017	**Norman H. Nachtrieb Memorial Award for Undergraduate Studies in Chemistry**	*$1,500*

Peer-Reviewed Publications

[1] **Csernica, T.,** Eiler, J. M., and Sessions, A. L. "Site-specific, Multi-Isotope Characterization of DDT Isolated from Sediment Samples." *Marine Pollution Bulletin.* **2026.**

[2] Wang, X., Liu, C., Han, J., Liu, J., **Csernica, T.,** Zhang, N., Ding, H., Gropp, J., McIntosh, J., Li, S., Xu, S., & Ellam, R. M. "Clumped isotopes tracing methane cycle and budget: a review." *Earth Science Reviews.* ***2026.***

[3] **Csernica, T.,** Moran, J. J., Fraga, C. G., and Eiler, J. M. "Simultaneous observation of 2H and 13C enrichment of methyl phosphonic acid via Orbitrap-IRMS with applications to nerve agent forensics." *Talanta.* **2025.**

[4] Wang, X., Biying, C., Chen, L, Dong, G., **Csernica, T.**, Zhang, N., Liu, J., Shuai, Y., Liu, C., Xu, Z., Li, S., and Xu, S. "Biogenic methane clumped isotope signatures: insights from microbially enhanced coal bed methane and machine learning." *Fuels.* **2024.**

[5] Ammanamanchi, N., Yester, J., Bargaje, A. P., Thomas, D., Little, K. C., Janzef, S., Francis, K., Weinberg, J., Johnson, J., Seery, T., Harris, T. H., Funari, B. J., Rose-Felker, K., Zinn, M., Miller, S. A., West, S. C., Feingold, B., Zhou, H., Steinhauser, M. L., **Csernica, T.,** Michener, R., and Kühn, B. "Elimination of ^{15}N-thymidine after oral administration in human infants." *PLOS One.* **2024.**

[6] **Csernica, T.**, Sessions, A. L., and Eiler, J. M. "High-Dimensional Isotomics Part 2: Observations of Over 100 Constraints on Methionine's Isotome." *Chemical Geology.* **2023.**

[7] Weiss, G. M., Sessions, A. L., Julien, M., **Csernica, T.,** Yamada, K., Gilbert, A., Freeman, K. H., and Eiler, J. M. "Intramolecular carbon isotope variation of labeled alanine." *International Journal of Mass Spectrometry.* **2023.**

[8] **Csernica, T.**, Bhattacharjee, S., and Eiler, J. M. "Accuracy and precision of ESI-Orbitrap-IRMS observations of hours to tens of hours via reservoir injection." *International Journal of Mass Spectrometry.* **2023.**

[9] **Csernica, T.**, and Eiler, J. M. "High-Dimensional Isotomics Part 1: A Mathematical Framework for Isotomics." *Chemical Geology.* **2023**.

Other Publications

[1] **Csernica, T., "**Extinction in single and multi-type branching processes." For University of Chicago Mathematics REU, **2015.**

Presentations

Invited Oral Presentations

[1] **Inaugural International Orbitrap-IRMS Users Meeting.** Goldschmidt Conference. **2025.** *Declined*

[2] **ThermoFisher Orbitrap Workshop,** American Geophysical Union Conference. "Data Quality Tests for Orbitrap-IRMS Data, Including Lessons from the 100 Isotopologue Challenge." **2023**.

[3] **ThermoFisher Orbitrap Workshop,** Orbitrap Isotope Users Meeting. "Expanding the Orbitrap-IRMS Frontier." **2023**.

[4] **Invited Talk,** American Society for Mass Spectrometry**. "**Mapping the Isotope Anatomies of Organic Molecules at Natural Abundance & Beyond". **2022**.

[5] **Invited Talk,** Yuk Yung Lunch Seminar. "Primitive Genetics and the Origin of Life." **2020.**

Oral Presentations

[1] **Northwestern Civil and Architectural Engineering Seminar.** "Getting the Most From AI in Your Research." *With Jiaxing Wang.* **2026**.

[2] **Chemical Forensics International Working Group.** "Isotopic Characterization of Sarin Precursors via Orbitrap Mass Spectrometry." **2026**.

[3] **Center for Synthetic Biology REU, Northwestern**. "Getting Started With AI Tools." *With Xave Bower.* **2026**.

[4] **American Chemical Society**. "Orbitrap Mass Spectrometry for Stable Isotope Forensics: A Case Study of Sarin Precursors." **2024**.

[5] **International Symposium on Isotopologues**. "High-Dimensional Isotomics: Observation and Interpretation of Over 100 Isotopic Constraints on Methionine." **2022**.

Poster Presentations

[1] **Goldschmidt Virtual**. "Reconstructing Extraterrestrial Organic Synthesis via Site-Specific Isotopic Measurements. **2021**.

[2] **Simons Collaboration on the Origins of Life Annual Meeting, Virtual**. "Isotopic Characterization of Nucleobases from a Possible Extraterrestrial Reaction Network. **2020**.

[3] **Gordon Conference on the Origin of Life**. "Site-specific Isotopic Fingerprinting of an Abiotically Plausible Adenine Synthesis. **2020**.

Teaching & Mentoring Experience

2026	Mentor, Center for Synthetic Biology REU	Northwestern
2019	Chemical Synthesis and Characterization for Chemical Engineering, TA	Caltech
2018-2019	General Chemistry, TA	Caltech

Teaching Workshops

Northwestern University Practicum: The Science of How Students Learn Spring 2026